

- [6] J. Certaine. The solution of ordinary differential equations with large time constants. *Mathematical methods for digital computers*, pages 128–132, 1960.
- [7] C. Clancy and J. A. Pudykiewicz. On the use of exponential time integration methods in atmospheric models. *Tellus A: Dynamic Meteorology and Oceanography*, 65(1):20898, Dec. 2013.
- [8] S. Cox and P. Matthews. Exponential time differencing for stiff systems. *Journal of Computational Physics*, 176(2):430–455, 2002.
- [9] N. Crouseilles, L. Einkemmer, and J. Massot. Exponential methods for solving hyperbolic problems with application to collisionless kinetic equations. *Journal of Computational Physics*, 420:109688, Nov. 2020.
- [10] D. Durran. *Numerical methods for fluid dynamics : with applications to geophysics*. Springer, New York, 2010.
- [11] ECMWF and P. White. IFS Documentation CY25R1 - Part III: Dynamics and Numerical Procedures. 2003.
- [12] J. Galewsky, R. K. Scott, and L. M. Polvani. An initial-value problem for testing numerical models of the global shallow-water equations. *Tellus A: Dynamic Meteorology and Oceanography*, 56(5):429–440, Jan. 2004.
- [13] S. Gaudreault and J. A. Pudykiewicz. An efficient exponential time integration method for the numerical solution of the shallow water equations on the sphere. *Journal of Computational Physics*, 322:827–848, Oct. 2016.
- [14] T. S. Haut, T. Babb, P. G. Martinsson, and B. A. Wingate. A high-order time-parallel scheme for solving wave propagation problems via the direct construction of an approximate time-evolution operator. *IMA Journal of Numerical Analysis*, 36(2):688–716, June 2015.
- [15] M. Hochbruck and A. Ostermann. Exponential integrators of rosenbrock-type. *Oberwolfach Reports*, 3:1107–1110, 2006.
- [16] M. Hochbruck and A. Ostermann. Exponential integrators. *Acta Numerica*, 19:209–286, May 2010.
- [17] M. Hortal. The development and testing of a new two-time-level semi-lagrangian scheme (settl5) in the ecmwf forecast model. *Quarterly Journal of the Royal Meteorological Society*, 128(583):1671–1687, July 2002.
- [18] C. Jablonowski and D. L. Williamson. The pros and cons of diffusion, filters and fixers in atmospheric general circulation models. In P. Lauritzen, C. Jablonowski, M. Taylor, and R. Nair, editors, *Numerical Techniques for Global Atmospheric Models*, chapter 13, pages 381–493. Springer Berlin Heidelberg, 2011.
- [19] J. D. Lawson. Generalized runge-kutta processes for stable systems with large lipschitz constants. *SIAM Journal on Numerical Analysis*, 4(3):372–380, Sept. 1967.
- [20] M. S. Longuet-Higgins. The eigenfunctions of laplace’s tidal equations over a sphere. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 262(1132):511–607, Feb. 1968.
- [21] V. T. Luan, J. A. Pudykiewicz, and D. R. Reynolds. Further development of efficient and accurate time integration schemes for meteorological models. *Journal of Computational Physics*, 376:817–837, Jan. 2019.
- [22] P. Lynch. A lagrange–laplace integration scheme for weather prediction and climate modelling. *Meteorology*, 1(4):355–376, Sept. 2022.
- [23] G. Mengaldo, A. Wyszogrodzki, M. Diamantakis, S.-J. Lock, F. X. Giraldo, and N. P. Wedi. Current and emerging time-integration strategies in global numerical weather and climate prediction. *Archives of Computational Methods in Engineering*, 26(3):663–684, Feb. 2018.
- [24] B. V. Minchev and W. M. Wright. A review of exponential integrators for first-order semi-linear problems. Technical report, Norwegian University of Science and Technology, 2005.
- [25] U. N. Oceanic and A. A. N. . N. C. for Environmental Prediction (NCEP). Global Forecast System - Global Spectral Model (GSM) - V13.0.2. <https://vlab.noaa.gov/web/gfs/documentation>, May 2016.
- [26] P. S. Peixoto and M. Schreiber. Semi-Lagrangian Exponential Integration with Application to the Rotating Shallow Water Equations. *SIAM Journal on Scientific Computing*, 41(5):B903–B928, 2019.
- [27] P. S. Peixoto, J. Thuburn, and M. J. Bell. Numerical instabilities of spherical shallow-water models considering small equivalent depths. *Quarterly Journal of the Royal Meteorological Society*, 144(710):156–171, Dec. 2017.
- [28] J. A. Pudykiewicz and C. Clancy. Convection experiments with the exponential time integration scheme. *Journal of Computational Physics*, 449:110803, Jan. 2022.
- [29] G. Rainwater, K. C. Viner, and P. A. Reinecke. Exponential time integration for 3d compressible atmospheric models, 2023.
- [30] Y. Saad. Variations on arnoldi’s method for computing eigenelements of large unsymmetric matrices. *Linear Algebra and its Applications*, 34:269–295, Dec. 1980.
- [31] N. Schaeffer. Efficient spherical harmonic transforms aimed at pseudospectral numerical simulations. *Geochemistry, Geophysics, Geosystems*, 14(3):751–758, Mar. 2013.
- [32] M. Schreiber and R. Loft. A parallel time integrator for solving the linearized shallow water equations on the rotating sphere. *Numerical Linear Algebra with Applications*, 26(2), Oct. 2018.

- [33] M. Schreiber, N. Schaeffer, and R. Loft. Exponential integrators with parallel-in-time rational approximations for the shallow-water equations on the rotating sphere. *Parallel Computing*, 85:56–65, July 2019.
- [34] R. K. Scott, L. M. Harris, and L. M. Polvani. A test case for the inviscid shallow-water equations on the sphere. *Quarterly Journal of the Royal Meteorological Society*, 142(694):488–495, Nov. 2015.
- [35] V. V. Shashkin and G. S. Goyman. Semi-lagrangian exponential time-integration method for the shallow water equations on the cubed sphere grid. *Russian Journal of Numerical Analysis and Mathematical Modelling*, 35(6):355–366, Dec. 2020.
- [36] W. C. Skamarock. Evaluating mesoscale nwp models using kinetic energy spectra. *Monthly Weather Review*, 132(12):3019–3032, Dec. 2004.
- [37] A. Staniforth and J. Côté. Semi-lagrangian integration schemes for atmospheric models—a review. *Monthly Weather Review*, 119(9):2206–2223, Sept. 1991.
- [38] J. Stoer and R. Bulirsch. *Introduction to numerical analysis*. Texts in Applied Mathematics. Springer, New York, NY, 3 edition, Aug. 2002.
- [39] P. N. Swarztrauber. Spectral transform methods for solving the shallow-water equations on the sphere. *Monthly Weather Review*, 124(4):730–744, Apr. 1996.
- [40] C. Temperton. Treatment of the coriolis terms in semi-lagrangian spectral models. *Atmosphere-Ocean*, 35(sup1):293–302, Jan. 1997.
- [41] J. Thuburn. Numerical wave propagation on the hexagonal c-grid. *Journal of Computational Physics*, 227(11):5836–5858, May 2008.
- [42] S. Vasylykevych and N. Žagar. A high-accuracy global prognostic model for the simulation of rossby and gravity wave dynamics. *Quarterly Journal of the Royal Meteorological Society*, 147(736):1989–2007, Mar. 2021.
- [43] D. L. WILLIAMSON. The evolution of dynamical cores for global atmospheric models. *Journal of the Meteorological Society of Japan. Ser. II*, 85B:241–269, 2007.
- [44] D. L. Williamson, J. B. Drake, J. J. Hack, R. Jakob, and P. N. Swarztrauber. A standard test set for numerical approximations to the shallow water equations in spherical geometry. *Journal of Computational Physics*, 102(1):211–224, Sept. 1992.